Cyclohexane Reactor Ver 1 · operator training simulator

Relief load basis

Document no. CHX-V1-PE-RLB-001 · **Discipline** Process engineering

Client Department of Chemical Engineering, Faculty of Engineering, Chulalongkorn University (teaching)

• 9 relief devices (PSV-101, -102, -104, -112, -109 and TSV-113, -106, -102J, -102F from the Safety-layers register SL-R8…R16) and EDV-101; 37 scenarios considered, the governing one sized with API 520 and a standard API 526 orifice.
• Orifices: PSV-101 **Q** · PSV-102 **K** · PSV-104 **J** · PSV-112 **J** · PSV-109 **P** · TSV-113 **E** · TSV-106 **D** · TSV-102J **E** · TSV-102F **D**; EDV-101 restriction orifice 1,756 mm².
• Flare: governing contingency Single: PSV-101 governing case (178.7 t/h) → unit header 30" (Mach 0.26).
• 11 model relief runs on copies of the simulator with every trip disabled; the model limits are recorded as engineering queries.
• Checks against the model and the other documents: 19 of 28 pass; each failure is one of the 10 engineering queries EQ-er-1…10.
• **Rev 1**: vessel elevations, PSV inlet and tailpipe lengths, the unit header length and the PSV-109 vent stack taken from the plot plan CHX-V1-PL-PP-001 (EQ-er-10 closed; answers EQ-la-2, -3, -8). See section 10.

Revisions

Rev	Date	Description	Originator	Checker	Approver
0	2026-10-26	Issued for teaching	Apinan Soottitantawat		
1	2026-10-27	Elevations and relief line lengths from the plot plan CHX-V1-PL-PP-001 (EQ-er-10 closed)	Apinan Soottitantawat		

This document is generated for teaching from the simulator model of the unit. It is not issued for construction; every number comes from the model or is labelled assumed / typical practice. No client, contractor or vendor is implied.

Contents

1 · Purpose, basis and method 4

1.1 Relieving conditions . . . 4

1.2 Sizing equations (API 520 Part I, SI) . . . 4

1.3 Fluid properties . . . 4

2 · Relief device summary 5

3 · Scenarios per device 6

PSV-101 · R-101 overhead (no isolation between it and R-101) · set 36 bar g → flare . . . 6

PSV-102 · R-102 off-gas line (R-102 and R-103 in one zone) · set 36 bar g → flare . . . 7

PSV-104 · V-104 HP separator (K-101 suction, E-106 process side, purge to HIC-107) · set 36 bar g → flare . . . 7

PSV-112 · K-101 discharge upstream of FV-103 · set 42 bar g → flare . . . 8

PSV-109 · Steam drum D-101 (with the E-101 shell, downcomer and riser) · set 8 bar g → atmosphere . . . 8

TSV-113 · E-103 reflux condenser, cooling-water side · set 10 bar g → drain . . . 9

TSV-106 · E-106 product cooler, cooling-water side · set 10 bar g → drain . . . 9

TSV-102J · R-102 jacket (cooling-water service, HS-102 COOL) · set 10 bar g → drain . . . 10

TSV-102F · Catalyst candle filters F-102A/B (one each) · set 36 bar g → closed drain . . . 10

4 · Inlet loss and back pressure 11

5 · Flare load summary and unit header 12

6 · Depressuring EDV-101 13

7 · Model relief runs 14

pv105 · PV-105 make-up valve wide open (PIC-105 output 100 %), make-up header at design . . . 14

pv105hi · PV-105 wide open with the make-up header at 45 bar g (assumed maximum supply pressure, as the SRS process-safety-time run) . . . 14

lv104 · V-104 liquid outlet blocked (LIC-104 output 0 %: LV-104 closed, its fail position) . . . 15

cw · Cooling water to E-103 lost (HV-113 shut) with feed and hydrogen running . . . 15

pa · Pump-around lost (P-101A stops, P-101B not started): E-101 removes nothing . . . 16

d101 · D-101 steam export blocked (PIC-110 output 0 %: PV-110 closed) at the design E-101 duty . . . 16

heat · HS-101 switched to HEAT with HV-110 open at operating temperature (MP steam on the E-101 shell) . . . 16

k101 · K-101 dead-headed: FV-103 closed (FIC-103 output 0 %) with K-101 running . . . 17

k101mu · K-101 dead-headed and PV-105 wide open (double failure), make-up header at design . . . 17

power · Total power failure: every motor stops (K-101, P-100A/B, P-101A/B, P-102A/B, M-101) . . . 17
edv · ESD with EDV-101 depressuring to flare (the model EDV-101 flow 0.12·√(P − 1.2) kmol/s) . . . 18
8 · Checks against the model and the other documents 19
9 · Engineering queries 21
10 · Assumptions 22

1 · Purpose, basis and method

This relief load basis (RLB) lists every pressure-relief device of Cyclohexane Reactor Ver 1, what each protects, every scenario of API 521 that can raise the pressure there, the relieving load of each, the governing case and the orifice selected with API 520 Part I. It is written from the simulator model (design state, noise off; and runs on copies of the model with the trips disabled), the Safety-layers register (devices and SL_ASSUMED set / design pressures), the line list CHX-V1-PE-LL-001, the P&ID CHX-V1-PID-001 and the HAZOP CHX-V1-SA-HZP-001 (recommendations R02, R13, R29, R30). Nothing is from a vendor or licensor; assumed values are listed in section 10.

1.1 Relieving conditions

Set pressure = the design pressure of the protected equipment (SL_ASSUMED: hydrogen loop 36 bar g, K-101 discharge 42 bar g, D-101 8 bar g, cooling-water sides 10 bar g). Allowable overpressure 10 % (single device) and 21 % (fire). Relieving pressure P1 = set × (1 + overpressure) + 1.013 bar: 40.61 bar a (loop, non-fire) and 44.57 bar a (loop, fire). API 521 rules applied: the control valves not involved in the failure stay at their normal position (no credit for a controller helping), trips and alarms are not credited, double jeopardy is not a design case unless stated.

1.2 Sizing equations (API 520 Part I, SI)

Gas / vapour, critical flow: $A = W / (C \cdot K_d \cdot P_1 \cdot K_b \cdot K_c) \cdot \sqrt{(T \cdot Z / M)}$, $C = 0.03948 \cdot \sqrt{(k \cdot (2/(k+1))^{(k+1)/(k-1)})}$; critical if $P_2 \le P_1 \cdot (2/(k+1))^{k/(k-1)}$, otherwise $A = W / (735 \cdot F_2 \cdot K_d \cdot K_c) \cdot \sqrt{(T \cdot Z / (M \cdot P_1 \cdot (P_1 - P_2)))}$. Steam: $A = 190.5 \cdot W / (P_1 \cdot K_d \cdot K_b \cdot K_c \cdot K_N \cdot K_{SH})$. Liquid: $A = 11.78 \cdot Q / (K_d \cdot K_w \cdot K_c \cdot K_v) \cdot \sqrt{(G / (P_1 - P_2))}$. A mm², W kg/h, P kPa, T K, Q L/min. K_d 0.975 (gas, steam) and 0.65 (liquid), K_c = 1 (typical preliminary values). Fire (API 521, wetted area within 7.6 m of grade, adequate drainage and fire-fighting): $Q = 43\,200 \cdot F \cdot A^{0.82}$ W, $W = Q / \lambda$. Thermal relief of blocked-in liquid: $q = \beta \cdot \Phi / (\rho \cdot c)$.

1.3 Fluid properties

Cyclohexane from the model correlations (PsatC, TbC, lamC; Watson T_c 280.35 °C at which PsatC = 40.7 bar a), vapour k 1.06, Z 1 (conservative); near the critical point a latent-heat floor of 115 kJ/kg is used (typical practice). Hydrogen-rich gas: design recycle purity 92.07 % (M 3.128), make-up M 2.086, k 1.4, Z 1.02. Steam and water: model PsatW/TsatW/lamW and typical saturation properties.

2 · Relief device summary

Device	Item	Service / protects	Set bar g	Design bar g	Governing scenario	Load kg/h	P1 bar a	T °C	A req mm²	Orifice	Valve flanges	Type	To
PSV-101	SL-R8	R-101 overhead (no isolation between it and R-101) — R-101, E-103 (process side), V-103, R-101 overhead line 10"-PG-2001-B2	36	36	Loss of cooling: Cooling-water failure: E-103 (and E-106) lose cooling, boil-up no longer condensed	178,724	40.6	255.5	5,297	Q (7,129.0)	6" × 8"	conventional	flare
PSV-102	SL-R9	R-102 off-gas line (R-102 and R-103 in one zone) — R-102, R-103, 8"-PG-2004-B2	36	36	Fire: External pool fire on the wetted surface of R-102 + R-103	39,778	44.6	280.1	959.0	K (1,185.8)	3" × 4"	conventional	flare
PSV-104	SL-R10	V-104 HP separator (K-101 suction, E-106 process side, purge to HIC-107) — V-104, E-106 (process side), K-101 suction, 8"-H-2007-B2	36	36	Control-valve failure: PV-105 wide open with the make-up header at its maximum 45 bar g (assumed)	5,646	40.6	44.9	657.8	J (830.3)	3" × 4"	conventional	flare
PSV-112	SL-R11	K-101 discharge upstream of FV-103 — K-101 discharge, 8"-H-2008-B2 to FV-103	42	42	Blocked outlet: Dead-head with the suction at the PSV-104 relieving pressure (make-up header 45 bar g, assumed)	8,513	47.2	60.0	713.0	J (830.3)	3" × 4"	conventional	flare
PSV-109	SL-R12	Steam drum D-101 (with the E-101 shell, downcomer and riser) — D-101, E-101 (shell / water side), 16"-S-3006-A1, 8"-BFW-3005-A1	8	8	Control-valve failure: MP steam on the E-101 shell (HS-101 HEAT, HV-110 open) with the shell connected to D-101	15,120	9.81	179.3	3,010	P (4,116.1)	4" × 6"	conventional	atmosphere
TSV-113	SL-R13	E-103 reflux condenser, cooling-water side — E-103 (CW side), 10"-CW-3007/3008-A1	10	10	Thermal expansion: Blocked-in cooling water heated by condensing vapour	8,074	12.0	—	78.50	E (126.5)	1" × 2"	conventional	drain
TSV-106	SL-R14	E-106 product cooler, cooling-water side — E-106 (CW side), 8"-CW-3009/3010-A1	10	10	Thermal expansion: Blocked-in cooling water heated by the reactor effluent	3,117	12.0	—	29.80	D (71.0)	1" × 2"	conventional	drain
TSV-102J	SL-R15	R-102 jacket (cooling-water service, HS-102 COOL) — R-102 jacket, 6"-CW-3011-A1	10	10	Thermal expansion: Jacket full of cooling water blocked in with R-102 at 160 °C	10,360	12.0	—	99.10	E (126.5)	1" × 2"	conventional	drain
TSV-102F	SL-R16	Catalyst candle filters F-102A/B (one each) — F-102A, F-102B	36	36	Thermal expansion: Filter isolated full of liquid for cleaning, warmed by the sun	4.670	40.6	—	0.0254	D (71.0)	1" × 2"	conventional	closed drain

Design pressures: SL_ASSUMED.Pdes and the other SL_ASSUMED values (Mechanical equipment list CHX-V1-ME-EQL-001 to confirm). Valve flanges: typical API 526 sizes for the orifice letter.

3 · Scenarios per device

PSV-101 · R-101 overhead (no isolation between it and R-101) · set 36 bar g → flare

Protects: R-101, E-103 (process side), V-103, R-101 overhead line 10"-PG-2001-B2. Normal operating pressure 30.00 bar g (PI-101). Register item SL-R8.

Scenario	Case	Status	P1 bar a	T °C	Fluid	Load kg/h	A req mm²	Basis
Blocked outlet	Gas outlet of R-101 blocked (valve closed in error toward R-102)	cannot lift	—	—	—	0	—	K-101 dead-head from the normal suction = Ps·(1 + kk) = 30.01 × 1.1538 − 1.013 = 33.62 bar g < set 36 bar g: the pressure source cannot lift the valve.
Control-valve failure	FV-101 benzene feed wide open	cannot lift	—	—	—	0	—	P-100 only delivers while PI-101 < 36 bar a = 34.99 bar g (model shut-off) < set 36: overfilling is a level hazard (LIC-101, R05), not an over-pressure case.
Control-valve failure	PV-105 make-up wide open	covered	—	—	—	0	—	The make-up enters at the K-101 suction; with K-101 running R-101 sits 1.0 bar above V-104 and PSV-101 lifts first, but the load is the same gas — sized on PSV-104 (where it enters and where it acts when K-101 is stopped).
Loss of cooling	Cooling-water failure: E-103 (and E-106) lose cooling, boil-up no longer condensed	GOVERNING	40.6	255.5	cyclohexane vapour + recycle gas	178,724	5,297	Cyclohexane vapour V = 7851 / 13596 kJ/kmol = 0.5774 kmol/s = 48.6 kg/s; with G1 (M 5.09) → W = 178724 kg/h, M = 63.35, k = 1.077
Fire	External pool fire on the wetted surface of R-101 + V-103	sized · near-critical	44.6	280.1	cyclohexane vapour	12,900	311.0	W = 412 / 115 × 3600 = 12900 kg/h
Runaway reaction	Runaway: all R-101 heat removal lost (E-101 and E-103) with hydrogen at FV-103 full open	flagged · two-phase	40.6	258.3	two-phase (vapour-only lower bound)	762,296	19,768	Vapour-only LOWER BOUND with the largest λ (at 258.3 °C, 154.5 kJ/kg): W ≥ 762296 kg/h. The real relief is two-phase (foaming slurry, dissolved H2): DIERS methods are outside the scope of this RLB.
Loss of cooling	Pump-around lost (E-101 removes nothing)	cannot lift	—	—	—	0	—	Model run: TI-100 peaks at 224 °C, below the boiling ceiling 258 °C; the extra boil-up (103 t/h) is condensed by E-103 (capacity 47.9 MW, line list) — no relief.
Utility failure	Total power failure	cannot lift	—	—	—	0	—	Model run: K-101 stops, no hydrogen reaches R-101 and the reaction stops; TI-100 falls to 166 °C — no relief (see the flare summary).

Governing case worked calculation (Loss of cooling)

- Heat removed by E-103 at design = V·(λ + cp·(T − Tc)) = 0.1567 kmol/s × (21230 + 200 × (200 − 55.7)) = 7851 kW (model terms): with the condenser lost it raises vapour instead
- Relieving pressure P1 = 36 × 1.10 + 1.013 = 40.61 bar a; pure cyclohexane would boil at TbC(P1) = 280.1 °C, at the model's critical point (280.4 °C / 40.7 bar a)
- The gas G1 = 0.2062 kmol/s keeps sweeping R-101, so the liquid boils where the model's stripping relation V = G1·Psat/(P1 − Psat) equals Q/λ(T): T = 255.5 °C, Psat 29.92 bar a, λ = 161.5 kJ/kg
- Cyclohexane vapour V = 7851 / 13596 kJ/kmol = 0.5774 kmol/s = 48.6 kg/s; with G1 (M 5.09) → W = 178724 kg/h, M = 63.35, k = 1.077

Required 5,297 mm² (critical flow) → selected **Q** 7,129.0 mm² (1.35 × required).

Model: Model run (HV-113 shut, trips off): reflux falls to 0, the vapour not refluxed rises to 54.1 t/h and is taken up by R-102 (TI-102 → 198.1 °C); PI-101 stays 29.9–30.05 bar g because the model's gas inventory holds no vapour (EQ-er-2). TIC-101 raises the E-101 duty in the model (R-101 settles at 205.3 °C); API 521 gives no credit for that.

Flag — Runaway reaction: two-phase relief; DIERS methods are outside the scope of this RLB; the load shown is a vapour-only lower bound and needs 19,768 mm², more than the selected orifice.

PSV-102 · R-102 off-gas line (R-102 and R-103 in one zone) · set 36 bar g → flare

Protects: R-102, R-103, 8"-PG-2004-B2. Normal operating pressure 29.60 bar g (PI-102). Register item SL-R9.

Scenario	Case	Status	P1 bar a	T °C	Fluid	Load kg/h	A req mm²	Basis
Blocked outlet	R-102 → R-103 → V-104 gas path blocked	cannot lift	—	—	—	0	—	Manual valves locked open (assumed); the only source is K-101 at ≤ 33.62 bar g (dead-head) < 36 bar g.
Control-valve failure	TV-102 wide open in HEAT (MP steam on the jacket)	cannot lift	—	—	—	0	—	MP steam condenses at 184 °C; R-102 boils at TbC(40.6 bar a) = 280.1 °C at the relieving pressure — steam cannot raise vapour there. Temperature (not pressure) concern: R03.
Loss of cooling	Jacket CW lost	cannot lift	—	—	—	0	—	No cooling duty in normal operation (the jacket heats); covered by the PSV-101 cooling-water case for the loop.
Fire	External pool fire on the wetted surface of R-102 + R-103	GOVERNING · near-critical	44.6	280.1	cyclohexane vapour	39,778	959.0	W = 1271 / 115 × 3600 = 39778 kg/h
Runaway reaction	R-102 reaction heat with no heat sink	covered · two-phase	40.6	280.1	cyclohexane vapour	15,900	420.7	r2·ΔH = 0.00247 × 206000 = 508 kW → 15900 kg/h at λ = 115 kJ/kg: smaller than the fire case.

Governing case worked calculation (Fire)

- R-102: liquid 1.52 m above the bottom tangent → wetted area 26.36 m² within 7.6 m of grade
- R-103: liquid 5.71 m above the bottom tangent → wetted area 35.43 m² within 7.6 m of grade (liquid above 7.6 m not counted)
- A = 61.8 m² → Q = 43 200 × 1 × 61.8^0.82 = 1270681 W = 1271 kW
- P1 = 36 × 1.21 + 1.013 = 44.57 bar a (21 % fire accumulation) > the model critical pressure 40.7 bar a: the liquid is supercritical; λ = 115 kJ/kg (floor, typical practice) at 280.1 °C
- W = 1271 / 115 × 3600 = 39778 kg/h

Required 959.0 mm² (critical flow) → selected **K** 1,185.8 mm² (1.24 × required).

PSV-104 · V-104 HP separator (K-101 suction, E-106 process side, purge to HIC-107) · set 36 bar g → flare

Protects: V-104, E-106 (process side), K-101 suction, 8"-H-2007-B2. Normal operating pressure 29.00 bar g (PI-105). Register item SL-R10.

Scenario	Case	Status	P1 bar a	T °C	Fluid	Load kg/h	A req mm²	Basis
Blocked outlet	Liquid outlet LV-104 closed (fail position)	cannot lift	—	—	—	0	—	Model run: LT-104S reaches 110 % (model cap) while PI-105 stays 29 bar g — the model keeps a fixed gas volume (EQ-er-2). Liquid can only be pushed by P-100, which stops at 34.99 bar g < 36: overfill to K-101 is handled by LSHH-104, not by relief.
Control-valve failure	PV-105 wide open, make-up header at design	cannot lift	—	—	—	0	—	Model run: PI-101 peaks at 34.5 bar g and PI-105 at 33.5 bar g (consumption + purge balance the make-up) < 36; even with no consumption the supply 36 bar a = 34.99 bar g cannot reach the set (no make-up flow at the relieving pressure, the supply being below it).
Control-valve failure	PV-105 wide open with the make-up header at its maximum 45 bar g (assumed)	GOVERNING	40.6	44.9	make-up hydrogen (99.5 % H2)	5,646	657.8	Model run with the header at 45 bar g (trips off): PI-101 crosses 36 bar g after 37.75 s, average rise 9.58 bar/min; net gas gain then 0.459 kmol/s with consumption credited.
Loss of cooling	E-106 cooling lost	covered	—	—	—	0	—	Cyclohexane vapour recirculates with the recycle gas; PIC-105 cuts the make-up; any loop accumulation is the PSV-101 cooling-water case.
Blocked outlet	Purge HIC-107 closed	cannot lift	—	—	—	0	—	No pressure source: purity falls, pressure is held by PIC-105.
Fire	External pool fire on the wetted surface of V-104	sized · near-critical	44.6	280.1	cyclohexane vapour	23,990	578.4	W = 766 / 115 × 3600 = 23990 kg/h

Governing case worked calculation (Control-valve failure)
• Model valve equation at the relieving pressure: Q = Cv105·√(Pmu − P1) = 0.323898 × √(46 − 40.61) = 0.7518 kmol/s (model cap muMax 0.9)
• W = 0.7518 × 2.086 × 3600 = 5646 kg/h at 44.9 °C (V-104)
• Model run with the header at 45 bar g (trips off): PI-101 crosses 36 bar g after 37.75 s, average rise 9.58 bar/min; net gas gain then 0.459 kmol/s with consumption credited.
Required 657.8 mm² (critical flow) → selected **J** 830.3 mm² (1.26 × required).
Model: At the set crossing the make-up runs at the model cap 0.9 kmol/s (FI-106 72621 Nm³/h) — a model limit, not the valve (EQ-er-3).

PSV-112 · K-101 discharge upstream of FV-103 · set 42 bar g → flare

Protects: K-101 discharge, 8"-H-2008-B2 to FV-103. Normal operating pressure 32.00 bar g (PI-112). Register item SL-R11.

Scenario	Case	Status	P1 bar a	T °C	Fluid	Load kg/h	A req mm²	Basis
Blocked outlet	K-101 dead-headed: FV-103 closed (fail closed)	cannot lift	—	—	—	0	—	Model run: PI-112 peaks at 36.54 bar g (= suction × (1 + kk) at zero flow, suction up to 31.53 bar g after the reaction stops) < 42.
Blocked outlet	Dead-head with PV-105 wide open, header at design	cannot lift	—	—	—	0	—	Model run: suction 34.98 bar g → PI-112 40.52 bar g < 42 (make-up supply 36 bar a limits the suction).
Blocked outlet	Dead-head with the suction at the PSV-104 relieving pressure (make-up header 45 bar g, assumed)	GOVERNING	47.2	60.0	recycle hydrogen	8,513	713.0	W = 0.7561 × 3.128 × 3600 = 8513 kg/h at 60 °C (K-101 discharge, recirculation heating not modelled)
Fire	Fire on K-101 discharge piping	cannot lift	—	—	—	0	—	No liquid inventory: gas-filled piping is protected by depressuring (EDV-101), not sized for fire (typical practice).

Governing case worked calculation (Blocked outlet)
• Model compressor curve Ph = Ps·(1 + kk·(1 − 0.35·(q/Qd)²)); suction Ps = 40.61 bar a (PSV-104 relieving)
• Dead-head Ph = 46.86 bar a > set 43.01 bar a; the curve reaches the set at q = Qd·√((1 − (Pset/Ps − 1)/kk)/0.35) = 0.7561 kmol/s
• W = 0.7561 × 3.128 × 3600 = 8513 kg/h at 60 °C (K-101 discharge, recirculation heating not modelled)
Required 713.0 mm² (critical flow) → selected **J** 830.3 mm² (1.16 × required).

PSV-109 · Steam drum D-101 (with the E-101 shell, downcomer and riser) · set 8 bar g → atmosphere

Protects: D-101, E-101 (shell / water side), 16"-S-3006-A1, 8"-BFW-3005-A1. Normal operating pressure 4.500 bar g (PI-110). Register item SL-R12.

Scenario	Case	Status	P1 bar a	T °C	Fluid	Load kg/h	A req mm²	Basis
Blocked outlet	Steam export blocked (PV-110 closed) at full E-101 duty	sized	9.81	179.3	saturated steam	12,628	2,514	P1 = 8 × 1.10 + 1.013 = 9.813 bar a (Tsat 179.3 °C); model accumulation 6.2 % < 10 %.
Control-valve failure	MP steam on the E-101 shell (HS-101 HEAT, HV-110 open) with the shell connected to D-101	GOVERNING	9.81	179.3	MP steam	15,120	3,010	The MP header (10 bar g) exceeds the set (8 bar g): PSV-109 must pass the full HV-110 flow (EQ-ep-3).
Control-valve failure	LV-109 wide open (fail open) with the export blocked	info	9.81	—	boiler feed water	13,832	144.5	Model LV-109: mb = Cv109·√(Pbfw − P1) = 3.52673 × √(11 − 9.81) = 3.842 kg/s of water (ρ 950 kg/m³, typical)
Tube rupture	E-101 tube rupture: hot slurry (≈31 bar g) into the shell / drum (4.5 bar g)	flagged · two-phase	9.81	181.5	cyclohexane flashing into water/steam (two-phase)	12,685	1,259	Flash at the shell relieving pressure: x = cp·(T − TbC(P1))/λ = 200 × (200 − 181.5) / 22965 = 0.161 → vapour-only 12.7 t/h (lower bound; hydrocarbon would vent to atmosphere through PSV-109)
Fire	Fire on the E-101 shell (D-101 above 7.6 m)	sized	10.7	183.0	steam	1,360	248.5	λ = lamW(183 °C) = 1968 kJ/kg → W = 1360 kg/h
Utility failure	Power failure	cannot lift	—	—	—	0	—	No hot slurry through E-101: no steam is raised (model run: PI-110 falls).

Governing case worked calculation (Control-valve failure)
• MP steam capacity of HV-110 = model cap msMax 4.2 kg/s = 15.1 t/h (line 8"-S-3002-A1 rated 15.1 t/h)
• The MP header (10 bar g) exceeds the set (8 bar g): PSV-109 must pass the full HV-110 flow (EQ-ep-3).
Required 3,010 mm² (critical flow) → selected **P** 4,116.1 mm² (1.37 × required).
Model: Model run (HEAT, HV-110 open at 200 °C): E-101 duty falls to 0 MW (steam at 184 °C cannot heat a 200 °C reactor) and D-101 loses its heat (PI-110 → 2.48 bar g): the model has no shell pressure and cannot show this case (EQ-er-4).
Flag — Tube rupture: two-phase relief; DIERS methods are outside the scope of this RLB; the load shown is a vapour-only lower bound.

TSV-113 · E-103 reflux condenser, cooling-water side · set 10 bar g → drain

Protects: E-103 (CW side), 10"-CW-3007/3008-A1. Normal operating pressure — bar g (CW). Register item SL-R13.

Scenario	Case	Status	P1 bar a	T °C	Fluid	Load kg/h	A req mm²	Basis
Thermal expansion	Blocked-in cooling water heated by condensing vapour	GOVERNING · boiling	12.0	—	cooling water	8,074	78.49	q = β·Φ/(ρ·c) = 1.27e-3 × 7850539 / (877 × 4453) = 153.38 L/min (typical water properties)
Tube rupture	E-103 tube rupture: loop gas (30 bar g) into the cooling-water side	flagged · two-phase	12.0	200.0	gas into water (two-phase)	4,616	548.8	Gas through 2 tube ends (Cd 0.6, choked) = 4616 kg/h; a thermal relief valve on a liquid-full CW side cannot pass it — gas reaches the CW return (R17).

Governing case worked calculation (Thermal expansion)
• Heat rate Φ = 7851 kW (E-103 design duty, model terms); water heats toward 200 °C, limited to Tsat at the relieving pressure 188.2 °C → 188.2 °C
• q = β·Φ/(ρ·c) = 1.27e-3 × 7850539 / (877 × 4453) = 153.38 L/min (typical water properties)
Required 78.49 mm² (critical flow) → selected **E** 126.5 mm² (1.61 × required).
Flag — Tube rupture: two-phase relief; DIERS methods are outside the scope of this RLB; the load shown is a vapour-only lower bound and needs 548.8 mm², more than the selected orifice.

TSV-106 · E-106 product cooler, cooling-water side · set 10 bar g → drain

Protects: E-106 (CW side), 8"-CW-3009/3010-A1. Normal operating pressure — bar g (CW). Register item SL-R14.

Scenario	Case	Status	P1 bar a	T °C	Fluid	Load kg/h	A req mm²	Basis
Thermal expansion	Blocked-in cooling water heated by the reactor effluent	GOVERNING	12.0	—	cooling water	3,117	29.79	q = β·Φ/(ρ·c) = 1.08e-3 × 3491670 / (908 × 4342) = 57.21 L/min (typical water properties)
Tube rupture	E-106 tube rupture: loop gas (29.3 bar g) into the cooling-water side	flagged · two-phase	12.0	158.6	gas into water (two-phase)	1,443	535.3	Gas through 2 tube ends (Cd 0.6, choked) = 1443 kg/h; a thermal relief valve on a liquid-full CW side cannot pass it — gas reaches the CW return (R17).

Governing case worked calculation (Thermal expansion)
• Heat rate Φ = 3492 kW (E-106 sensible duty F3·cp·(T R-103 out − T V-104), model terms); water heats toward 158.6 °C, limited to Tsat at the relieving pressure 188.2 °C → 158.6 °C
• q = β·Φ/(ρ·c) = 1.08e-3 × 3491670 / (908 × 4342) = 57.21 L/min (typical water properties)
Required 29.79 mm² (critical flow) → selected **D** 71.0 mm² (2.38 × required).
Flag — Tube rupture: two-phase relief; DIERS methods are outside the scope of this RLB; the load shown is a vapour-only lower bound and needs 535.3 mm², more than the selected orifice.

TSV-102J · R-102 jacket (cooling-water service, HS-102 COOL) · set 10 bar g → drain

Protects: R-102 jacket, 6"-CW-3011-A1. Normal operating pressure — bar g (jacket). Register item SL-R15.

Scenario	Case	Status	P1 bar a	T °C	Fluid	Load kg/h	A req mm²	Basis
Thermal expansion	Jacket full of cooling water blocked in with R-102 at 160 °C	GOVERNING	12.0	—	cooling water	10,360	99.11	q = β·Φ/(ρ·c) = 1.09e-3 × 11520002 / (907 × 4347) = 190.45 L/min (typical water properties)
Control-valve failure	MP steam (10 bar g) on the jacket in HEAT	flagged · set	—	—	—	0	—	The jacket runs on MP steam at 10 bar g in HEAT: the TSV set 10 bar g equals the steam pressure and would lift in normal service (EQ-ep-4).

Governing case worked calculation (Thermal expansion)
• Heat rate Φ = 11520 kW (UAj·(T R-102 − T CW) = 90 × (160 − 32), model); water heats toward 160 °C, limited to Tsat at the relieving pressure 188.2 °C → 160 °C
• q = β·Φ/(ρ·c) = 1.09e-3 × 11520002 / (907 × 4347) = 190.45 L/min (typical water properties)
Required 99.11 mm² (critical flow) → selected **E** 126.5 mm² (1.28 × required).

TSV-102F · Catalyst candle filters F-102A/B (one each) · set 36 bar g → closed drain

Protects: F-102A, F-102B. Normal operating pressure 32.60 bar g (P-102 discharge). Register item SL-R16.

Scenario	Case	Status	P1 bar a	T °C	Fluid	Load kg/h	A req mm²	Basis
Thermal expansion	Filter isolated full of liquid for cleaning, warmed by the sun	GOVERNING	40.6	—	cyclohexane	4.670	0.0254	Φ = solar 1000 W/m² × 2 m² = 2000 W (assumed); q = β·Φ/(ρ·c) = 0.0012 × 2000 / (779 × 1850) = 0.0999 L/min — the smallest standard valve is ample
Operating margin	Normal operation	cannot lift	—	—	—	0	—	F-102 runs at PI-102 + pump 3 bar ≈ 32.6 bar g = 91 % of the set (EQ-ep-5).

Governing case worked calculation (Thermal expansion)
• Φ = solar 1000 W/m² × 2 m² = 2000 W (assumed); q = β·Φ/(ρ·c) = 0.0012 × 2000 / (779 × 1850) = 0.0999 L/min — the smallest standard valve is ample
Required 0.0254 mm² (critical flow) → selected **D** 71.0 mm² (2,795.28 × required).

4 · Inlet loss and back pressure

Inlet line: valve inlet size (enlarged until the non-recoverable loss at the rated flow is ≤ 3 % of set), 1.5 m where the valve sits on the vessel nozzle (assumed); PSV-104 5.3 m, PSV-112 6.1 m as routed on the plot plan, ΣK 0.8 (assumed), Darcy fully rough. Outlet: isothermal compressible flow in the tailpipe (PSV-101 20.5 m, PSV-102 17.8 m, PSV-104 19 m, PSV-112 6.4 m to the unit header (plot plan route lengths, used as equivalent lengths); PSV-109 vent stack 9.9 m to atmosphere), started at the outlet flange size or the P&ID size and enlarged until the total back pressure is ≤ 10 % of set (conventional valve) and the exit Mach ≤ 0.7. Superimposed back pressure = the unit-header inlet pressure of the governing flare contingency (section 5). TSVs discharge liquid to an atmospheric drain.

Device	Rated kg/h	Inlet NPS	v m/s	ΔP inlet bar	% of set (≤ 3)	Tailpipe NPS (P&ID)	Superimposed bar a	Built-up bar	Total % of set	Exit Mach	Valve type
PSV-101	240,546	6"	61.2	1.038	2.88	24" (6")	1.312	0.093	1.1	0.50	conventional
PSV-102	49,185	3"	35.1	0.570	1.58	10" (8")	1.312	0.167	1.3	0.47	conventional
PSV-104	7,127	3"	132.1	0.543	1.51	8" (8")	1.312	0.287	1.6	0.45	conventional
PSV-112	9,914	3"	110.5	0.688	1.64	8" (8")	1.312	0.159	1.1	0.52	conventional
PSV-109	20,673	6" (from 4")	62.3	0.091	1.13	8"	1.013	0.335	4.2	0.69	conventional
TSV-113	13,012	2" (from 1")	1.9	0.021	0.21	liquid to an atmospheric drain: negligible back pressure	—	0.000	0.0	—	conventional
TSV-106	7,429	1"	4.9	0.242	2.42	liquid to an atmospheric drain: negligible back pressure	—	0.000	0.0	—	conventional
TSV-102J	13,223	2" (from 1")	1.9	0.021	0.21	liquid to an atmospheric drain: negligible back pressure	—	0.000	0.0	—	conventional
TSV-102F	13,055	1"	10.0	0.871	2.42	liquid to an atmospheric drain: negligible back pressure	—	0.000	0.0	—	conventional

5 · Flare load summary and unit header

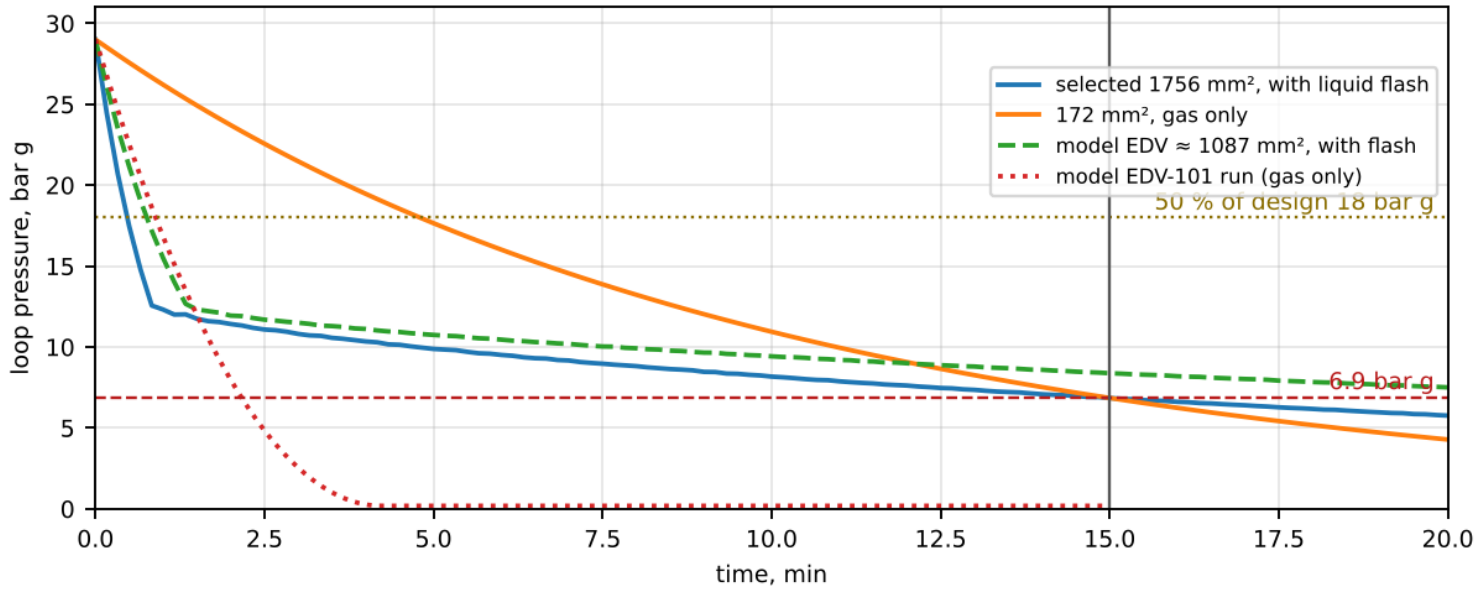
The header carries the largest load that arrives at the same time: each single failure, each global utility failure (cooling water, power, instrument air — loads they cause together) and one fire zone. Unit header from the last PSV to the knock-out drum V-190: 176 m equivalent, V-190 at 1.2 bar a (assumed); smallest size with Mach ≤ 0.5 and an inlet pressure ≤ the line list's superimposed back pressure 1.5 bar a: **30"** (inlet 1.312 bar a, Mach 0.26, 69 m/s). PSV-109 and the TSVs do not discharge to the flare.

Contingency	Devices	Load kg/h	M	T °C	Header inlet bar a	Mach	Note
Single: PSV-101 governing case	PSV-101	178,724	63.35	256	1.312	0.26	Cooling-water failure: E-103 (and E-106) lose cooling, boil-up no longer condensed
Single: PSV-102 governing case	PSV-102	39,778	84.16	280	1.204	0.05	External pool fire on the wetted surface of R-102 + R-103
Single: PSV-104 governing case	PSV-104	5,646	2.09	45	1.202	0.03	PV-105 wide open with the make-up header at its maximum 45 bar g (assumed)
Single: PSV-112 governing case	PSV-112	8,513	3.13	60	1.203	0.04	Dead-head with the suction at the PSV-104 relieving pressure (make-up header 45 bar g, assumed)
Global: cooling-water failure (E-103, E-106, R-102 jacket)	PSV-101	178,724	63.35	256	1.312	0.26	One vapour source (R-101 boil-up) — PSV-104 E-106 case is covered by it
Global: power failure (all motors)	—	0	—	—	1.200	—	Model run: K-101 stops → no hydrogen to R-101 → no heat release; D-101 receives no heat: no relief
Global: instrument-air failure (every control valve to its fail position)	—	0	—	—	1.200	—	PV-105 and FV-103 fail closed, FV-104 fails open (full cooling), PV-110 fails open: no valve opens a pressure source — no relief
Fire zone: reactor structure (R-101, R-102, R-103 together, assumed one zone)	PSV-101 + PSV-102	52,678	84.16	280	1.208	0.07	Fire zone extent assumed; EDV-101 normally opened in a fire (not added: it stops the PSVs lifting)
EDV-101 depressuring (ESD), initial rate with the selected orifice	EDV-101	11,718	3.13	60	1.206	0.05	Gas-and-flash basis; falls quickly

6 · Depressuring EDV-101

Criterion (API 521 guidance): reduce the loop to 6.9 bar g (or to 50 % of the design gauge pressure, 18 bar g) within 15 min. The loop gas of the model — K.Vloop 80 m³ at 29.00 bar g and 60 °C (the model's RTV corresponds to 59.9 °C), 92.07 % H_2, rest CH_4, 86.7 kmol = 271 kg — blows down adiabatically through a restriction orifice (C_d 0.85, back pressure 1.5 bar a; critical / subcritical orifice equation; the remaining gas expands isentropically, the cold lower bound). With the flash option the R-101 and R-102 liquid (R-101 209 kmol at 200 °C, R-102 114 kmol at 160 °C) boils whenever the pressure is below its vapour pressure and follows TbC(P), adding cyclohexane vapour.

Basis	Orifice for 6.9 bar g	Orifice for 18 bar g	6.9 bar g reached	Initial flow	Min. gas T
gas only	172 mm²	55 mm²	15.0 min	0.32 kg/s	−69 °C
with liquid flash (selected)	1,756 mm² (≈ 47.3 mm bore)	55 mm²	15.0 min	3.25 kg/s	−55 °C
model EDV-101 (≈ 1,087 mm²), gas only	—	—	130 s (model run)	2.02 kg/s	—
model EDV-101 with flash	—	—	not in 20 min	—	−40 °C



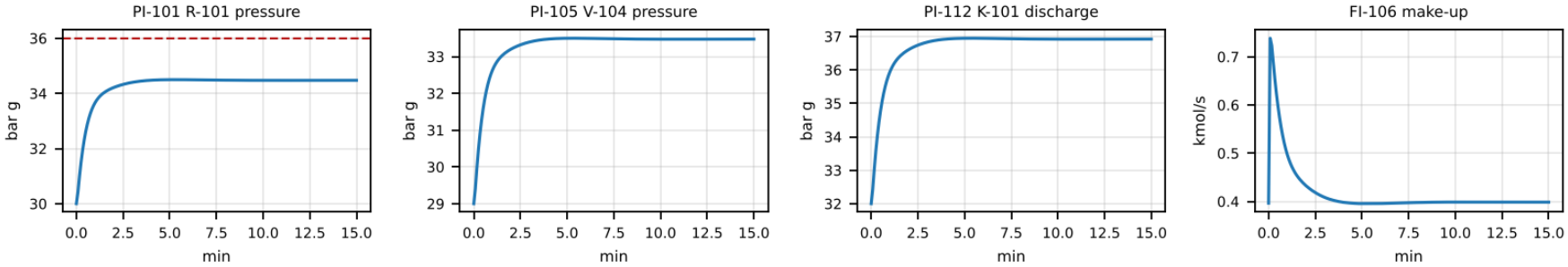
Flashed cyclohexane with the selected orifice: 59.8 kmol. The flash only starts below the R-101 vapour pressure, so the 50 %-of-design target needs the same orifice with or without it. The isentropic gas temperature is a lower bound for the minimum design metal temperature check (HAZOP R29); the walls warm the gas in reality.

7 · Model relief runs

Each run starts from the model design state on a copy of the simulator. For the relief cases every trip is disabled (K.T1hh, T3hh, P1hh, L4hh out of reach and the trip function stubbed, restored afterwards); the ESD run keeps the trips because ESD is the trip. The model has no relief valves on the hydrogen loop: the runs show how fast and how far pressure moves and what feeds it.

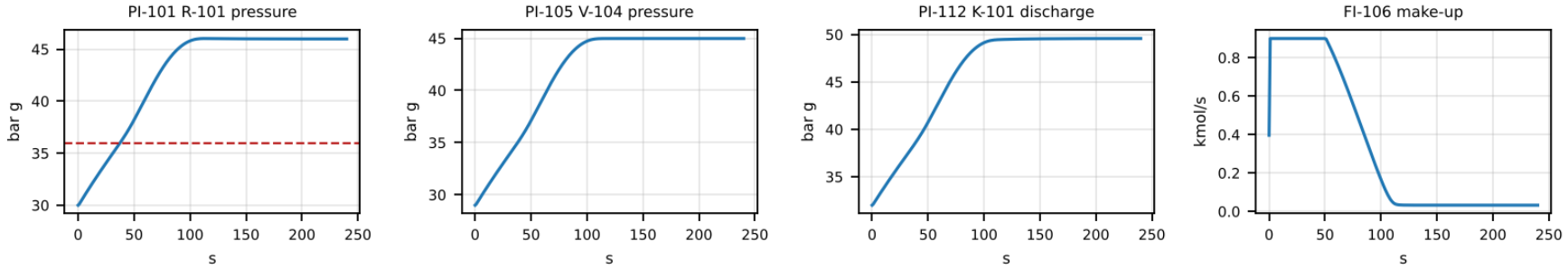
pv105 · PV-105 make-up valve wide open (PIC-105 output 100 %), make-up header at design

Devices PSV-104, PSV-101. Basis: K.Pmu = 36 bar a (model make-up supply). Result: peak 34.50 bar g against the set 36.



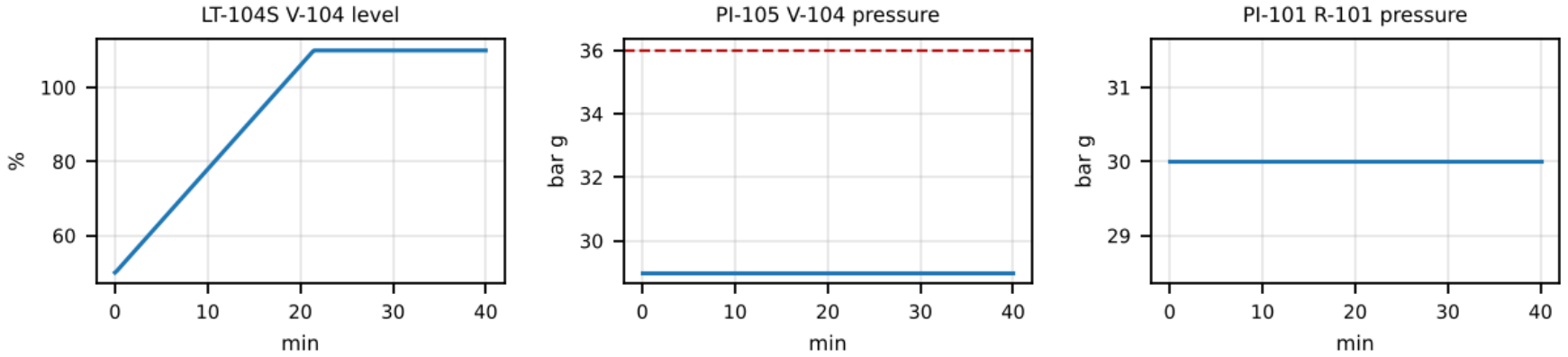
pv105hi · PV-105 wide open with the make-up header at 45 bar g (assumed maximum supply pressure, as the SRS process-safety-time run)

Devices PSV-104, PSV-101. Basis: K.Pmu raised to 46 bar a on the copy (assumed); restored in finally. Result: crosses the set 36 bar g at 37.8 s (average 9.58 bar g/min).



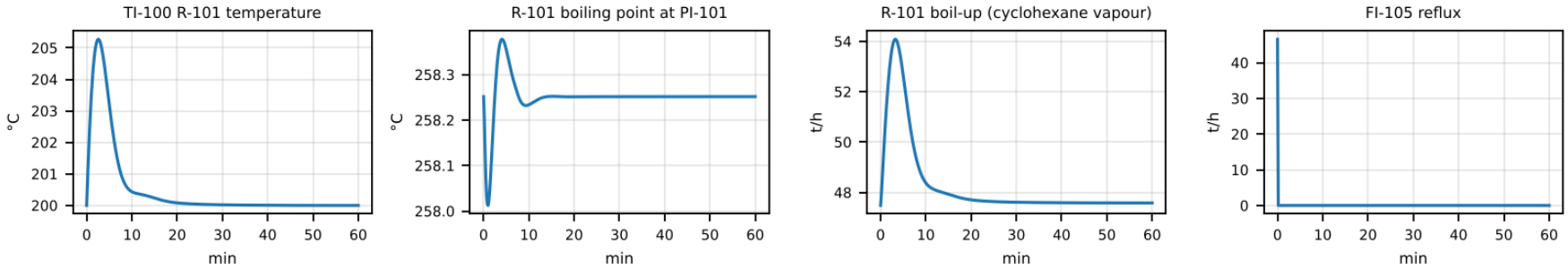
lv104 · V-104 liquid outlet blocked (LIC-104 output 0 %: LV-104 closed, its fail position)

Devices PSV-104. Basis: design state; trips disabled (LSHH-104 does not stop K-101). Result: peak 29.00 bar g against the set 36.



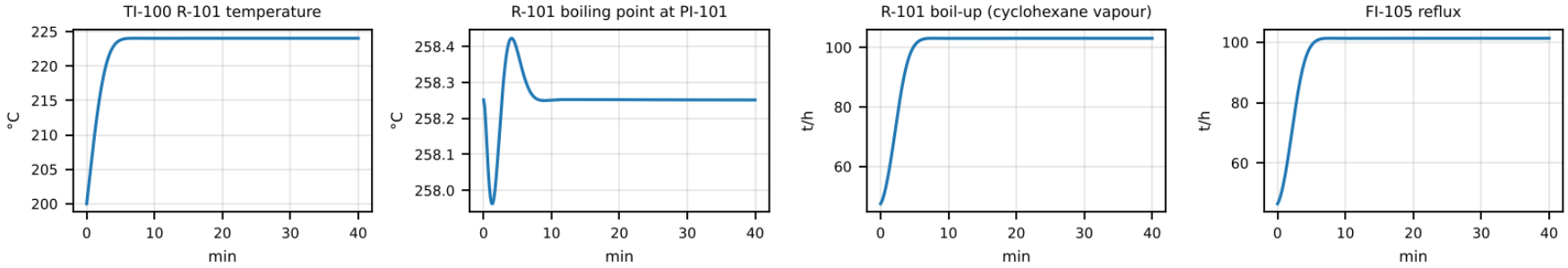
cw · Cooling water to E-103 lost (HV-113 shut) with feed and hydrogen running

Devices PSV-101. Basis: design state; trips disabled. Result: peak 30.05 bar g against the set 36.



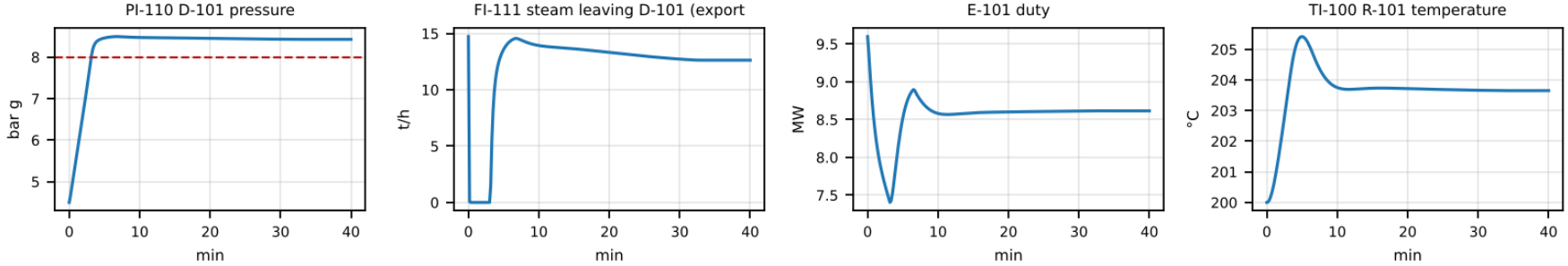
pa · Pump-around lost (P-101A stops, P-101B not started): E-101 removes nothing

Devices PSV-101. Basis: design state; trips disabled. Result: peak 30.07 bar g against the set 36.



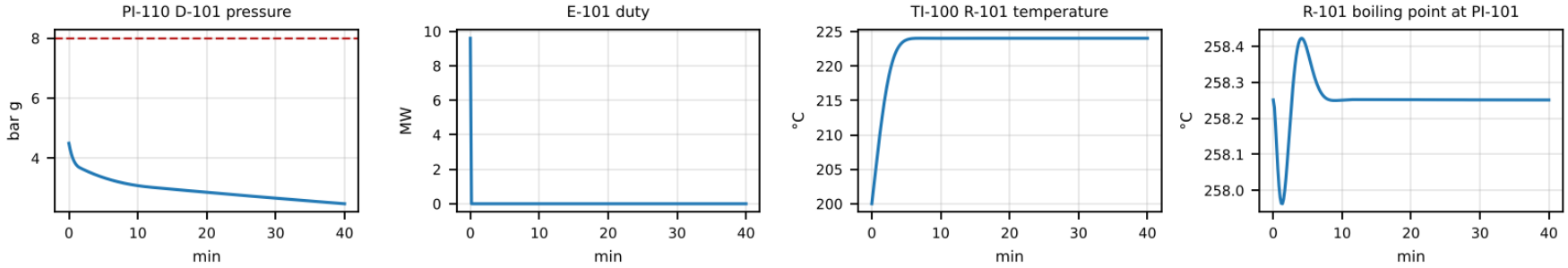
d101 · D-101 steam export blocked (PIC-110 output 0 %: PV-110 closed) at the design E-101 duty

Devices PSV-109. Basis: design state; trips disabled; the model releases steam above 9.0 bar a (ms += 8·(Pd − 9) kg/s), the analogue of PSV-109. Result: crosses the set 8 bar g at 188.0 s (average 1.12 bar g/min).



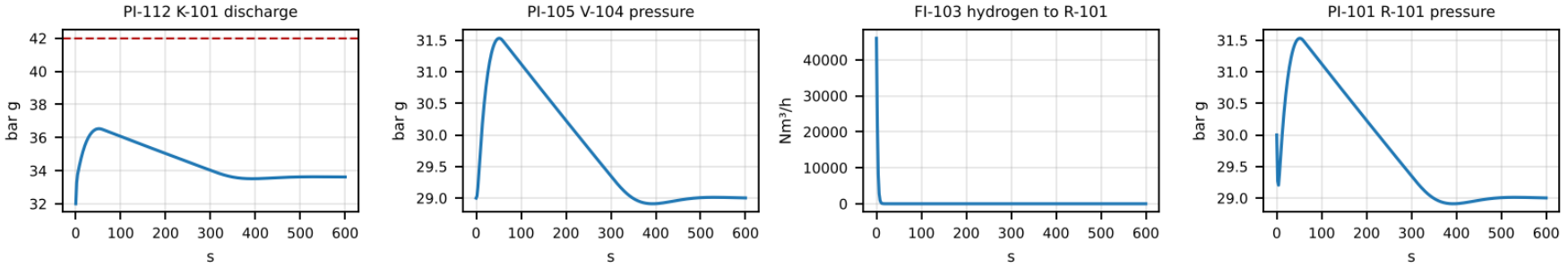
heat · HS-101 switched to HEAT with HV-110 open at operating temperature (MP steam on the E-101 shell)

Devices PSV-109. Basis: design state; trips disabled; in HEAT the model takes D-101 off E-101 (Qd = 0) and has no shell pressure. Result: peak 4.50 bar g against the set 8.



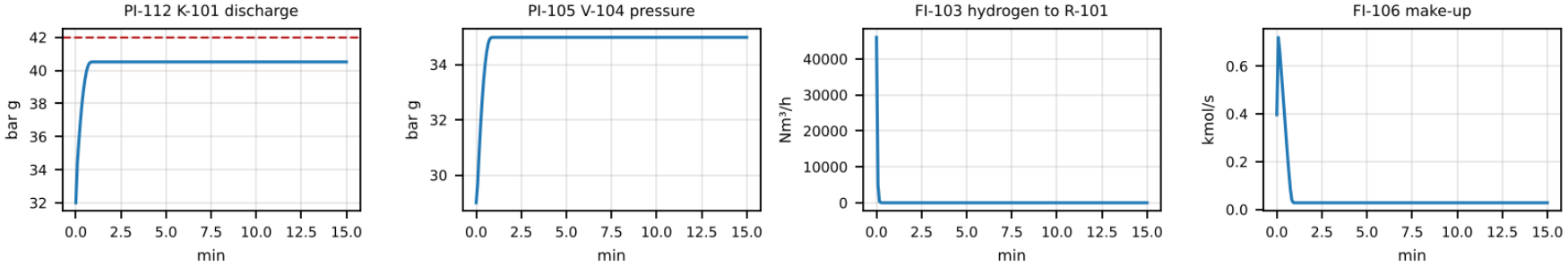
k101 · K-101 dead-headed: FV-103 closed (FIC-103 output 0 %) with K-101 running

Devices PSV-112. Basis: design state; trips disabled; PIC-105 in AUTO. Result: peak 36.54 bar g against the set 42.



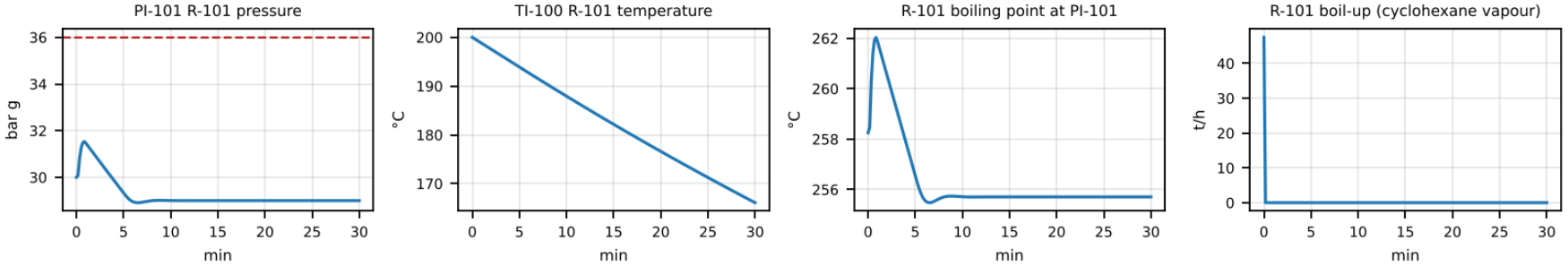
k101mu · K-101 dead-headed and PV-105 wide open (double failure), make-up header at design

Devices PSV-112. Basis: design state; trips disabled; K.Pmu = 36 bar a. Result: peak 40.52 bar g against the set 42.



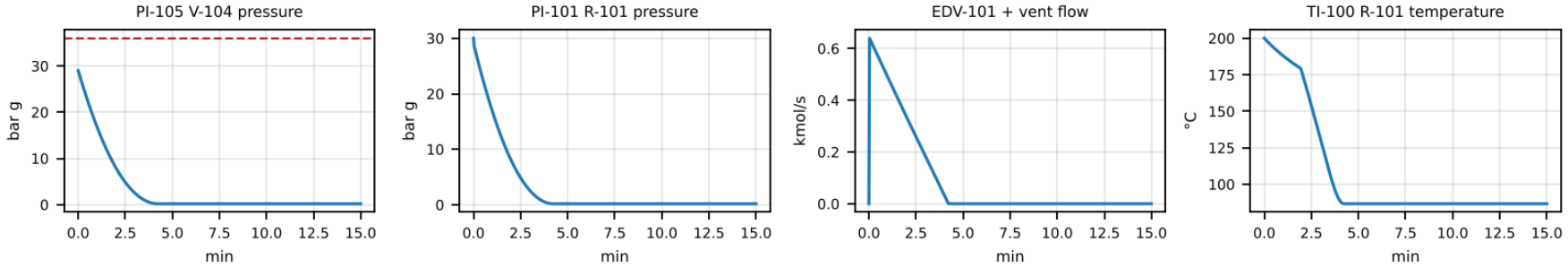
power · Total power failure: every motor stops (K-101, P-100A/B, P-101A/B, P-102A/B, M-101)

Devices flare. Basis: design state; trips disabled; CW and steam supplies kept (they come from outside the unit). Result: peak 31.54 bar g against the set 36.



edv · ESD with EDV-101 depressuring to flare (the model EDV-101 flow 0.12·√(P − 1.2) kmol/s)

Devices EDV-101. Basis: design state; trips ENABLED (ESD is the trip itself): XV-101, XV-105 shut, K-101 stopped, EDV-101 open. Result: peak 29.00 bar g against the set 36.



8 · Checks against the model and the other documents

Computed by the same function the tab runs (erCheckRows): 19 of 28 pass. A failed check is a finding, recorded as an engineering query.

#	Check	Compared	Result	OK
1	Every PSV / TSV of the Safety-layers register is in the RLB (and no other)	SL_IT (9) vs RLB (9)	PSV-101, PSV-102, PSV-104, PSV-109, PSV-112, TSV-102F, TSV-102J, TSV-106, TSV-113	✓
2	Set pressure in the RLB = setting in the register	SL_IT[tag] setting	all 9 equal	✓
3	Each set pressure ≤ design pressure of the protected equipment	EQL (Mechanical) where listed, else SL_ASSUMED	PSV-101 36/36 · PSV-102 36/36 · PSV-104 36/36 · PSV-112 42/42 · PSV-109 8/8 · TSV-113 10/10 · TSV-106 10/10 · TSV-102J 10/10 · TSV-102F 36/40	✓
4	Required areas recomputed = document	erGas / erSteam / erLiq on every row	worst difference 0.027 %	✓
5	Selected orifice ≥ required area (governing case)	API 526 area vs A req	PSV-101 Q 1.35× · PSV-102 K 1.24× · PSV-104 J 1.26× · PSV-112 J 1.16× · PSV-109 P 1.37× · TSV-113 E 1.61× · TSV-106 D 2.38× · TSV-102J E 1.28× · TSV-102F D 2,795.28×	✓
6	Flagged two-phase cases fit the selected orifice	vapour-only lower bounds	exceed: PSV-101 runaway reaction; TSV-113 tube rupture; TSV-106 tube rupture	✗ EQ-er-5, EQ-er-6
7	Inlet loss ≤ 3 % of set at rated flow (recomputed)	erInlet, inlet lines of part 5	PSV-101 2.9 % · PSV-102 1.6 % · PSV-104 1.5 % · PSV-112 1.6 % · PSV-109 1.1 % · TSV-113 0.2 % · TSV-106 2.4 % · TSV-102J 0.2 % · TSV-102F 2.4 %	✓
8	Back pressure within the valve type (≤ 10 % conventional, ≤ 50 % balanced; recomputed)	erIso tailpipes, header pressure	PSV-101 1.1 % · PSV-102 1.3 % · PSV-104 1.6 % · PSV-112 1.1 % · PSV-109 4.2 %	✓
9	PSHH-101 trips below the loop set (SIS acts before relief)	K.P1hh vs SL_ASSUMED.Pdes	33.5 < 36 bar g	✓
10	Make-up at the design supply cannot lift the loop PSVs	K.Pmu − 1.013 and run pv105 peak vs set	34.99 bar g supply; PI-101 peak 34.50 < 36	✓
11	K-101 dead-head at the highest design suction < PSV-112 set	K.Pmu·(1 + K.kk) − 1.013	40.52 < 42 bar g	✓
12	P-100 shut-off (model: feed only while P1 < 36 bar a) < loop set	36 − 1.013 bar g	34.99 < 36	✓
13	The model's D-101 release matches PSV-109 (opens at the set, ≤ 10 % accumulation)	model release above 9.0 bar a; run d101	7.99 bar g vs set 8; accumulation 6.2 %	✓
14	Operating pressure ≤ 90 % of set (typical margin for spring valves)	design state vs set	PSV-101 83 % · PSV-102 82 % · PSV-104 81 % · PSV-112 76 % · PSV-109 56 % · TSV-102F 91 %	✗ EQ-ep-5
15	MP steam that can reach the E-101 shell ≤ PSV-109 set	PsatW(K.Tms) − 1.013 vs SL_ASSUMED.Psv109	9.92 bar g (saturation at 184 °C; header 10 bar g nominal) vs 8	✗ EQ-ep-3, EQ-er-4
16	MP steam on the R-102 jacket in HEAT ≤ 90 % of the TSV-102J set	PsatW(K.Tms) − 1.013 vs SL_ASSUMED.Tsv	9.92 bar g = 99 % of 10	✗ EQ-ep-4
17	Tube rupture not credible: low side ≥ 10/13 of high side — E-101 (shell / drum)	API 521 10/13 rule	8 vs 30.8 bar g	✗ EQ-er-5
18	Tube rupture not credible: low side ≥ 10/13 of high side — E-103 (CW side)	API 521 10/13 rule	10 vs 27.7 bar g	✗ EQ-er-5
19	Tube rupture not credible: low side ≥ 10/13 of high side — E-106 (CW side)	API 521 10/13 rule	10 vs 27.7 bar g	✗ EQ-er-5
20	P&ID tailpipe size ≥ valve outlet	CHX-V1-PID-001 (HOLD lines) vs part 5	PSV-101 6" / 8" · PSV-102 8" / 4" · PSV-104 8" / 4" · PSV-112 8" / 4"	✗ EQ-er-9
21	Every device is drawn on the EPC P&ID sheets	CHX-V1-PID-001 sheet text	all 9 found	✓
22	Lines protected by a PSV have design pressure ≥ its set	CHX-V1-PE-LL-001 lines	all lines OK	✓

#	Check	Compared	Result	OK
23	HAZOP actions on relief answered by this RLB	HZ_REC R02 (loop relief), R13 (D-101), R29 (EDV), R30 (flare KO)	R02, R13, R29, R30 → parts 2, 2, 7, 6	✓
24	RLB design state = the model design state	design meas vs document	PI101 Δ0.000 · PI105 Δ0.000 · PI112 Δ0.000 · PI110 Δ0.000	✓
25	Flare header: Mach ≤ 0.5 and inlet ≤ the line list back pressure	governing contingency, 30"	Mach 0.26, 1.312 bar a ≤ 1.5	✓
26	EDV-101 orifice reaches 6.9 bar g in 15 min with the liquid flash (recomputed)	erBlow, 1,756 mm²	6.85 bar g at 15 min	✓
27	The model's EDV-101 reaches 6.9 bar g in 15 min once the liquid flash is counted	model-equivalent 1,087 mm², with flash	8.39 bar g at 15 min (model run, gas only: 130 s)	✗ EQ-er-7
28	Model relief run re-run (PV-105 open, 45 bar g header, trips disabled)	set crossing time vs embedded run	37.75 s vs 37.75 s	✓

9 · Engineering queries

ID	Query	Documents	Proposed resolution
EQ-er-1	Loop relief sits at the critical point of cyclohexane With the loop PSVs set at 36 bar g the relieving pressure is 40.61 bar a (10 %) and 44.57 bar a (fire, 21 %); the model's own correlations put the cyclohexane critical point at 280.4 °C / 40.7 bar a (PsatC at the Watson Tc of lamC). A blocked-in vessel in a fire is therefore supercritical and the fire loads use the 115 kJ/kg floor (typical practice) — uncertain. The cooling-water case stays below the critical point only because the recycle gas keeps sweeping R-101 (relieving at 255.5 °C); it needs Q for 178.7 t/h and sets the unit flare header.	CHX-V1-PE-RLB-001, SL_ASSUMED (Safety layers), CHX-V1-ME-EQL-001	Do a dynamic (supercritical-fluid) fire relief study; evaluate an SIS trip of feed and K-101 on loss of CW (HAZOP R06) as a high-integrity alternative under API 521, which would take the cooling-water case off the flare.
EQ-er-2	The model holds no vapour or liquid volume in the loop pressure Loop pressure is Ps = nG·RTV (permanent gas only, fixed 80 m³). Boil-up that is not refluxed becomes R-102 liquid (cooling-water run: vapour not refluxed 54 t/h, PI-101 unchanged), and a rising V-104 level does not compress the gas (blocked-outlet run: level 110 %, PI-105 unchanged). The model cannot show the over-pressure of these cases.	model.js step(), CHX-V1-PE-RLB-001 runs cw, lv104	Keep the hand calculation as the basis; for teaching, add a condensable-vapour term to the loop inventory and subtract the liquid volumes from K.Vloop (lead / model owner).
EQ-er-3	Model caps the make-up at muMax The make-up flow is capped at K.muMax = 0.9 kmol/s (72621 Nm³/h) whatever PV-105 and the header pressure allow. With the header at 45 bar g (assumed) the cap is active when PI-101 crosses the set (run pv105hi), so the run's rise rate is a model limit. The RLB load uses the valve equation at the relieving pressure (0.752 kmol/s, below the cap).	model.js K.muMax, SRS PST run, CHX-V1-PE-RLB-001	Confirm the maximum make-up header pressure (supplier) and the PV-105 Cv on its datasheet; size PSV-104 on that pressure.
EQ-er-4	E-101 shell in HEAT: MP steam 10 bar g against an 8 bar g relief (and no shell pressure in the model) HS-101 HEAT puts MP steam (10 bar g) on the E-101 shell. The equipment list designs the E-101 shell for 12 bar g but D-101 for 8 bar g: if the shell stays open to D-101, PSV-109 (set 8 bar g) must pass the full HV-110 flow (15.1 t/h) — this case governs PSV-109 (P). If the shell is isolated from D-101 in HEAT, the shell has no relief device at all. The model takes D-101 off E-101 in HEAT and has no shell pressure, so the run cannot show it.	EQ-ep-3, SL-R12, CHX-V1-PE-LL-001 8"-S-3002-A1, CHX-V1-ME-EQL-001 E-101 / D-101	Design D-101 and its circuit for ≥ 12 bar g with PSV-109 re-set, or put a pressure-reducing valve plus a PSV on 8"-S-3002-A1; give the E-101 shell its own PSV if it can be isolated from D-101; interlock HEAT with XV-101 (R16).
EQ-er-5	Exchanger tube rupture is credible on E-101, E-103 and E-106 Low-pressure sides (E-101 shell 8, CW sides 10 bar g) are below 10/13 of the tube-side design pressure (40 / 36 bar g). E-101: 78.9 t/h of hot slurry flashing (16 % vapour) into the steam system, vented to atmosphere through PSV-109. E-103 / E-106: loop gas into the cooling water that the thermal relief valves cannot pass.	CHX-V1-PE-RLB-001, CHX-V1-ME-EQL-001, HAZOP R15, R17	Raise the low-pressure-side design pressure to ≥ 10/13 of the high side, or add rupture-disc relief on the CW returns to a safe location and route PSV-109 away from grade; detect hydrocarbon in the D-101 blowdown and CW returns (R15, R17).
EQ-er-6	Runaway relief is two-phase and outside this basis Loss of both heat sinks with hydrogen at FV-103 full open releases 32.7 MW; the vapour-only lower bound is 762 t/h (2 × T), far above the selected Q. The real relief of a foaming slurry with dissolved hydrogen is two-phase; DIERS methods are outside the scope.	CHX-V1-PE-RLB-001, LOPA L1, SRS SIF-1	Treat the double failure through the SIS (TSHH-100, recommended LSLL/TDA items) and LOPA L1; if it stays a relief case, commission a DIERS two-phase study.
EQ-er-7	Liquid flash dominates EDV-101 depressuring; model EDV ignores it To reach 6.9 bar g in 15 min the loop gas alone needs 172 mm²; the R-101 liquid flashing below Psat(200 °C) = 12.4 bar g adds 59.8 kmol of cyclohexane and raises the orifice to 1756 mm². The model's EDV-101 (equivalent 1087 mm², gas only) reaches 6.9 bar g in 130 s; with the flash counted the same orifice is still at 8.4 bar g after 15 min. The isentropic gas temperature falls to -54.6 °C (cold lower bound).	HAZOP R29, model.js Qvent, CHX-V1-PE-LL-001 8"-FL-3013-A2	Size the EDV-101 restriction orifice on the flash basis; check minimum design metal temperature of the loop (R29) and the flare header for the cold gas.
EQ-er-8	V-104 size: PFD drawing smaller than the model level span The model's V-104 level span is 30 m³ (50 % = 15 m³ of liquid) while the PFD vessel 2.0 × 5.0 m holds 17.8 m³. The equipment list sizes V-104 2.3 × 9.1 m; this RLB uses that size for the fire case (wetted 33.4 m²).	model.js K.V4span, CHX-V1-PFD-001, CHX-V1-ME-EQL-001	Revise the PFD drawing size of V-104 to the equipment list; keep K.V4span consistent.
ID	**Query**	**Documents**	**Proposed resolution**
EQ-er-9	Tailpipe sizes on the P&ID (HOLD) against the selected valves PSV-101: valve outlet 8", P&ID 6", RLB 24" (built-up + superimposed 1.1 % of set, exit Mach 0.5); PSV-102: valve outlet 4", P&ID 8", RLB 10" (built-up + superimposed 1.3 % of set, exit Mach 0.47); PSV-104: valve outlet 4", P&ID 8", RLB 8" (built-up + superimposed 1.6 % of set, exit Mach 0.45); PSV-112: valve outlet 4",	CHX-V1-PID-001 6"-FL-1040, 8"-FL-2016 (HOLD), CHX-V1-PE-RLB-001	Release the HOLDs with the sizes of part 5 (one tailpipe per valve; the shared 8"-FL-2016 becomes a sub-header), and add the unit flare header to the line list.

ID	Query	Documents	Proposed resolution
	P&ID 8", RLB 8" (built-up + superimposed 1.1 % of set, exit Mach 0.52). The tailpipes were enlarged until the total back pressure is ≤ 10 % of set and the exit Mach ≤ 0.7 (typical practice); PSV-101 needs a 24" tailpipe because of the 179 t/h cooling-water case (EQ-er-1).		
EQ-er-10 CLOSED	Equipment elevations and relief line lengths: now from the plot plan (closed) The fire cases depend on the height of each vessel above grade (7.6 m rule). This revision reads the plot plan CHX-V1-PL-PP-001 (tabs/lay/layout.json): bottoms at R-101 7.95 m (was 4), R-102 5.8 m (was 2), R-103 2.49 m (was 1.5), V-103 15.3 m (was 9), V-104 1.45 m (horizontal drum as on the EQL and the plot plan; was vertical at 1), D-101 8.4 m, E-101 shell 0.98 m. Fire loads: PSV-101 (R-101) 12.9 t/h, PSV-102 (R-102 + R-103) 39.8 t/h, PSV-104 (V-104) 24 t/h. Lengths: tailpipes PSV-101 20.5 m, PSV-102 17.8 m, PSV-104 19 m, PSV-112 6.4 m (were 20 m); inlets PSV-104 5.3 m and PSV-112 6.1 m (were 1.5 m); unit header to V-190 176 m (was 150 m); PSV-109 vent stack 9.9 m (was 4 m). Result: orifices PSV-101 Q, PSV-102 K, PSV-104 J, PSV-112 J, PSV-109 P; inlet losses ≤ 2.9 % of set; total back pressure ≤ 4.2 % of set; unit header 30" (inlet 1.312 bar a, Mach 0.26). Still assumed: the F-102 projected area for solar heating.	CHX-V1-PL-PP-001 (EQ-la-2, EQ-la-3, EQ-la-8), CHX-V1-ME-EQL-001, CHX-V1-PE-RLB-001	Closed in this revision of the RLB (layout.json elevations and lengths). The plot plan copy of the RLB (layout.json src.rlb) refreshes when the Layout tab re-runs la_data.js; F-102 projected area stays assumed.

10 · Assumptions

Item	Value	Basis
Overpressure	10 % single, 21 % fire	API 520 / 521
K_d	0.975 gas/steam, 0.65 liquid; K_c 1	typical preliminary (certified values govern)
Latent heat floor	115 kJ/kg	typical practice near the critical point
Make-up header maximum	45.0 bar g	assumed (as the SRS process-safety-time run); model supply 35.0 bar g
Exchanger tube	19.05 × 14.83 mm, C_d 0.6	typical 3/4" 14 BWG
E-101 tube side	PI-101 + 1 bar	assumed
Inlet line	1.5 m where the valve sits on the vessel nozzle (assumed); PSV-104 5.3 m, PSV-112 6.1 m as routed on the plot plan; ΣK 0.8	plot plan CHX-V1-PL-PP-001 where routed; otherwise assumed
Tailpipes	PSV-101 20.5 m, PSV-102 17.8 m, PSV-104 19 m, PSV-112 6.4 m to the unit header (plot plan route lengths, used as equivalent lengths); PSV-109 vent stack 9.9 m; exit Mach ≤ 0.7	plot plan CHX-V1-PL-PP-001 (were 20 m and 4 m assumed); Mach typical
Unit flare header	176 m to V-190 at 1.2 bar a, Mach ≤ 0.5	route length from the plot plan (was 150 m = line list); KO pressure assumed
Thermal relief	water properties along saturation; F-102 solar 1000 W/m² × 2 m², cyclohexane β 0.0012 1/K	typical / assumed
EDV-101	C_d 0.85, back pressure 1.5 bar a, 6.9 bar g in 15 min	typical; API 521 guidance
Geometry R-101	3.8 × 4.5 m, vertical, bottom 7.95 m	EQL ID 3,800 × T/T 4,500 mm; bottom tangent EL +7.95 m (plot plan CHX-V1-PL-PP-001; was 4 m assumed)
Geometry R-102	3.2 × 3.8 m, vertical, bottom 5.8 m	EQL ID 3,200 × T/T 3,800 mm; bottom tangent EL +5.80 m (plot plan CHX-V1-PL-PP-001; was 2 m assumed)
Geometry R-103	1.95 × 7.4 m, vertical, bottom 2.49 m	EQL ID 1,950 × T/T 7,400 mm; bed 5.71 m; trickle bed wetted over the bed (typical practice); bottom tangent EL +2.49 m (plot plan CHX-V1-PL-PP-001; was 1.5 m assumed)
Geometry V-103	1.2 × 3 m, vertical, bottom 15.3 m	EQL ID 1,200 × T/T 3,000 mm (PFD size); bottom tangent EL +15.30 m (plot plan CHX-V1-PL-PP-001; was 9 m assumed)
Geometry V-104	2.3 × 9.1 m, horizontal, bottom 1.45 m	EQL ID 2,300 × T/T 9,100 mm; shell bottom (CL − D/2) EL +1.45 m (plot plan CHX-V1-PL-PP-001; was 1 m assumed)

Item	Value	Basis
Geometry D-101	2.4 × 8.4 m, horizontal, bottom 8.4 m	EQL ID 2,400 × T/T 8,400 mm; shell bottom (CL − D/2) EL +8.40 m (plot plan CHX-V1-PL-PP-001; was 8 m assumed)
Geometry E-101 shell	1.25 × 7.32 m, horizontal, bottom 0.98 m	EQL Shell ID 1,250 mm × 1210 tubes Ø25.4 × 7 320 mm; shell bottom (CL − D/2) EL +0.98 m (plot plan CHX-V1-PL-PP-001; was 1 m assumed)